DATA ANALYTICS PORTFOLIO PROJECT

UPI Transaction Analytics: Turning raw data into operational strategy.

A case study on structuring 500,000+ records into observation, prioritization, and action.



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CONTEXT

Structuring an observation layer for 500,000+ transactions

500,000+

Analyzed half a million records to evaluate system health, app market share, and transaction drop-off points.

Focused on building a self-service Excel dashboard, enabling stakeholders to track KPIs and spot bottlenecks without relying on expensive BI infrastructure.



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DATA FOUNDATION

I set the analytical foundation: clean first, interpret second

01

Timestamp

I standardized timestamps and fixed invalid date-syntax errors.

02

Source dataset

I removed null and blank values to create a stable source dataset.

03

INR ₹

I ensured currency consistency in INR ₹.

04

04. Data Validation

Cross-checked status and app fields to ensure data integrity before KPI calculation.

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FEATURE ENGINEERING

I used feature engineering to answer “when do transactions run into problems?”

I derived time-based slices from timestamps to examine load and stress over time.

01

Hour of Day

I extracted Hour of Day to view load by hour.

02

Day of Week

I extracted Day of Week to compare transaction rhythm across days.

03

Peak stress points

I combined the hour-by-status matrix to identify peak stress points.

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PEAK WINDOW

I prioritized monitoring the 18:00–21:00 peak window

Traffic was concentrated in the evening, between 18:00 and 21:00.

Concentrated volume

My hour-by-hour analysis showed high load in the evening window.

Co-occurring failure signals

The peak coincided with rising latency and more status drop-offs.

Operational implication

Capacity planning should be checked hour by hour, not only at daily totals.



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RISK BY APP

Apps carry very different operational risk

PhonePe: Monitored success vs. technical decline rates.

Google Pay: Balanced high volume tracking with transaction quality.

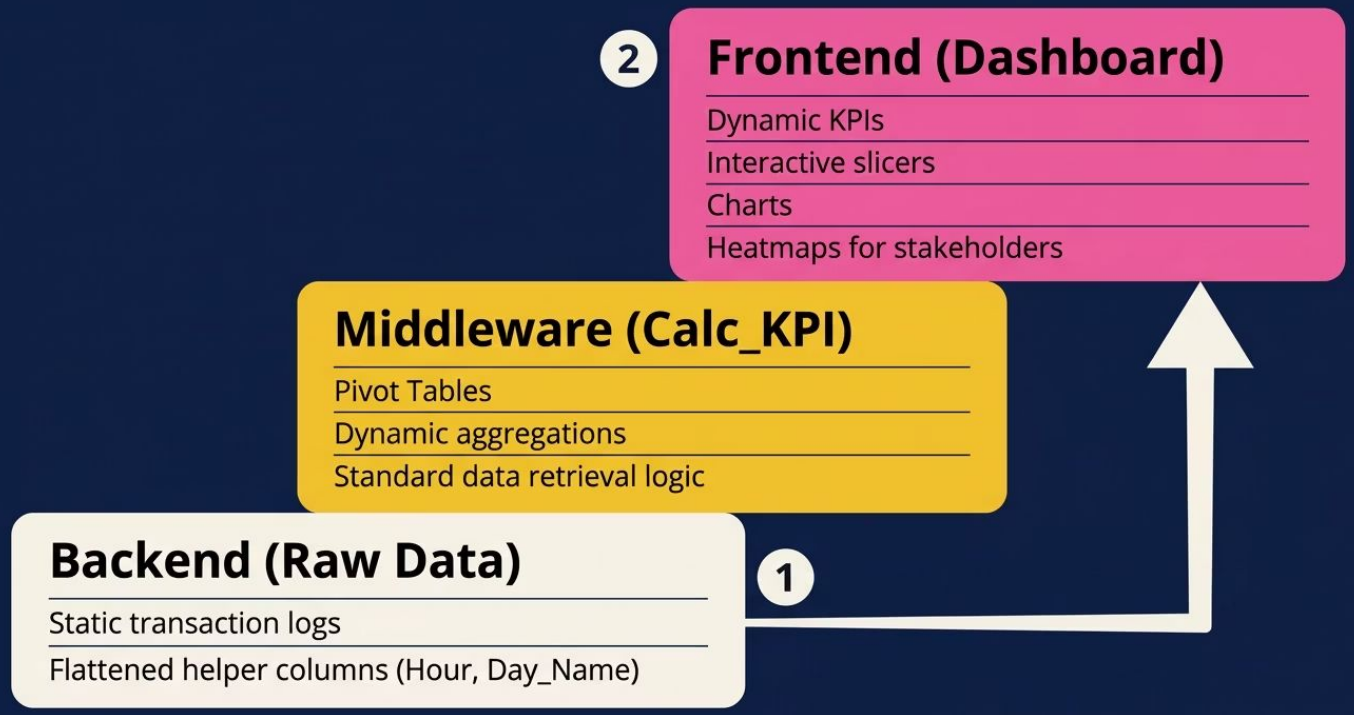
BHIM/Amazon Pay: Contextualized failure rates independent of market share.

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DATA PRODUCT

I designed the dashboard as a three-layer data product

Each layer has a clear role: storage, calculation, then delivering insight to stakeholders.



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FROM INSIGHT TO ACTION

From insight to action: reducing failures in peak hours first

01

Load balancing

Adjust capacity for the 18:00–21:00 window.

02

Partner focus

Prioritize apps showing high technical decline.

03

Monitoring

Track volume, success rate, latency and drop-off together.

04

Continuous Iteration

Turn the dashboard into a loop of detect, validate and improve.

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KEY TAKEAWAY

FROM INSIGHT TO IMPACT

- **What:** Revealed true performance. Shifted focus from vanity metrics (market share) to operational metrics (success rates vs. technical declines).
- **When & Where:** Pinpointed systemic bottlenecks. Identified the 18:00–21:00 window as the critical stress point causing latency and drop-offs across specific states.
- **What next:** Enabled targeted action. Provided a framework for proactive server load balancing during peak hours and targeted technical support for partner apps with high failure rates.

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